
PHAST: Grey-Box Port-Hamiltonian State-Space Models with Recoverable Physical Structure

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Abstract

1 Forecasting is not identification: a model can predict trajectories while assigning
2 the physical explanation to the wrong channel. This ambiguity is acute when
3 observations are limited to position or proprioceptive histories, because the history
4 must reveal hidden phase variables, the geometry that converts stored energy into
5 motion, and the dissipative channels through which energy is lost. We introduce
6 PHAST, a grey-box port-Hamiltonian state-space model that treats forecasting as
7 latent phase-state inference followed by typed physical evolution. PHAST first
8 maps observations through optional domain charts, infers a latent phase state from
9 history, and then evolves it with a port-Hamiltonian transition whose storage, mass
10 geometry, dissipation, and ports remain explicit components. This gives a practical
11 interface for knowledge injection: users may fix known physics, bound or structure
12 partially known components, or learn missing storage, geometry, dissipation, and
13 port terms without changing the rollout code path. Across thirteen position-only
14 benchmarks spanning mechanical, electrical, molecular, thermal, gravitational,
15 and ecological systems, PHAST is the best row-wise model against Transformer,
16 GRU, LSTM, S5, LinOSS, D-LinOSS, and VPT baselines, with gains from $1.5 \times$
17 to 4.5×10^5 while using 3.4k–13.7k trainable parameters. Beyond rollout error,
18 the experiments expose a specific identification failure: unconstrained dissipation
19 can act as an error sink, yielding accurate forecasts while recovering physically
20 meaningless parameters. We therefore evaluate both forecasting stability and
21 physical recovery, showing that bounded PSD damping improves attribution and
22 exposes when low rollout error hides the wrong dissipative mechanism.

23 1 Introduction

24 Physical systems can look unrelated at the surface: a pendulum slows under air drag, an RLC circuit
25 dissipates charge as heat, and a robot arm loses energy through joint friction. The port-Hamiltonian
26 view treats these as instances of the same accounting principle: energy-like quantities are stored,
27 exchanged, dissipated, and sometimes injected through ports. The goal of PHAST is to learn this
28 accounting from data, to forecast while recovering the interpretable physical structure that drives the
29 motion. This is where forecasting and identification diverge: a low-error trajectory can still hide the
30 wrong physical explanation if storage, geometry, or dissipation absorbs mismatch through the wrong
31 channel.

32 This is not merely a sequence prediction problem. Robots and scientific instruments often expose only
33 partial state: joint angles without momenta, charges without fluxes, positions without velocities, or
34 noisy proprioceptive histories rather than clean canonical coordinates. A single observed generalized
35 position, or configuration, q_t is generally not Markovian. Its recent history carries hidden information
36 about velocity, momentum, inertia, friction, damping, load, and environmental forcing. The modeling

question is therefore not only “what is the next q ?”, but “what latent physical state and physical channels explain this motion?” PHAST addresses this as a constrained forward-learning problem: it does not train a separate inverse flow from observations to physics, but uses an observer/canonicalizer to infer the phase state needed by one structured forward operator.

Problem setting. We formalize this as *position-only* (q-only) forecasting. We observe generalized positions q_t at discrete times; momenta p_t , port powers, and dissipative forces are latent. For robotics, this is the proprioceptive setting in which joint-position histories must stand in for full mechanical state. Given a length- K burn-in window $q_{t-K+1:t}$, the model must predict $\hat{q}_{t+1:t+H}$ over a horizon H and, when partial physics is available, recover interpretable components such as potential, mass geometry, damping, and port-compatible work. The main benchmark uses strict q-only supervision: models train from position histories and next-position targets. Appendix B separately tests a phase-supervised q-only regime where simulator momenta/flows are auxiliary training targets but deployment still consumes only q -histories.

PHAST rollout contract. PHAST treats q-only forecasting as neural state-space modeling:

$$\begin{aligned}\hat{q}_t &= o_\phi(q_{t-K+1:t}), \\ \hat{x}_t &= C_\psi(q_t, \hat{q}_t), \\ \hat{x}_{t+h+1} &= \Phi_{\Delta t}(\hat{x}_{t+h}; V_\theta, M_\theta, D_\theta), \quad \hat{q}_{t+h} = \Pi_q(\hat{x}_{t+h}).\end{aligned}\tag{1}$$

Here o_ϕ is a causal observer, C_ψ canonicalizes the inferred velocity-like state into phase variables, $\Phi_{\Delta t}$ is a structured physical transition, and Π_q reads out positions. The separation is deliberate: the observer/canonicalizer performs state inference from proprioceptive history, while the transition performs physical evolution in latent phase space.

Design requirements. The architecture is built to satisfy five requirements implied by the q-only setting. First, a history encoder must infer the latent phase state; feeding q_t alone to a Markov transition is generally insufficient. Second, the transition must expose typed physical channels—storage, conservative exchange, dissipation, and external ports—rather than only a hidden recurrent state. Third, domain knowledge must enter through both components and coordinates: a user may know gravity but not friction, capacitance but not effective resistance, an inertia template but not the dissipative law, or a target manifold such as positivity, periodicity, rotations, or SPD structure. Fourth, the dissipative channel must be positive semidefinite by construction and spectrally bounded when the goal includes physical recovery. Fifth, evaluation must distinguish stable forecasts from correct physical attribution. These requirements directly determine the components in Sec. 3: the observer handles hidden state, the pH core handles typed energy accounting, the definite parameterizations enforce admissible geometry and dissipation, and the experiments report both rollout and recovery metrics.

Contributions. These requirements lead to four design choices and one evaluation choice: **(1) Proprioceptive port-Hamiltonian state-space modeling:** a causal observer and canonicalizer infer a latent phase state from q-only history, and a port-Hamiltonian transition evolves that state through the conservative–dissipative decomposition; **(2) typed knowledge injection:** the same rollout operator covers KNOWN, PARTIAL, and UNKNOWN regimes by allowing $V_\theta(q)$, $M_\theta(q)$, $D_\theta(q)$, ports, and domain charts to be fixed, partially specified, bounded, or learned independently; **(3) efficient definite parameterizations:** low-rank outer-product expansions guarantee $D_\theta(q) \succeq 0$ and $M_\theta(q) \succ 0$ by construction while keeping damping and mass-vector products inexpensive; **(4) damping-spectrum control for identifiability:** bounded damping mitigates the error-sink failure mode where dissipation absorbs model mismatch and turns the damping block into an attribution channel rather than a generic stabilizer; and **(5) two-axis q-only evaluation:** forecasting metrics are separated from physical-recovery metrics across thirteen systems and seven sequence-model baselines. The rest of the paper follows this dependency order: related work identifies why conservative geometry and black-box sequence modeling are insufficient; the method then builds the observer, pH transition, damping parameterization, and integrator; the experiments finally separate “predicts well” from “recovers the right physics.”

2 Related work

Conservative and dissipative physical learners. Hamiltonian Neural Networks [Greydanus et al., 2019], Lagrangian Neural Networks [Cranmer et al., 2020], and DeLaN [Lutter et al., 2019] make energy or variational structure learnable. SympNets and volume-preserving transformers [Jin et al., 2020, Brantner et al., 2024] preserve conservative geometry. These priors are valuable when the relevant phase variables are available or can be reliably reconstructed, but strict conservation is not the right certificate for systems whose defining behavior is energy loss. Dissipative SymODEN [Zhong et al., 2020], DHNN [Sosanya and Greydanus, 2022], and port-Hamiltonian neural models [Desai et al., 2021, Eidnes and Riemer-Sørensen, 2023] add dissipation and ports. PHAST is closest to this line, but targets a harder interface: the model receives only q histories, must infer a phase state, and must keep learned damping from becoming an unconstrained error sink.

Latent dynamics and system-identification baselines. Neural ODEs [Chen et al., 2018] and latent ODE/ODE-RNN models [Rubanova et al., 2019] learn continuous-time latent transitions from sequences. They test whether a history encoder plus generic continuous dynamics is enough, but their latent coordinates do not by themselves expose storage, geometry, dissipated power, or port work. Classical identification tools such as SINDy [Brunton et al., 2016], DMD with control [Proctor et al., 2016], and deep Koopman embeddings [Lusch et al., 2018] are strong comparators when the representation, derivative estimates, or control template are favorable. PHAST instead fixes the accounting interface and lets users choose how much physics to supply: known templates, bounded grey-box components, or fully learned terms.

Efficient sequence models. GRU/LSTM, S5 [Smith et al., 2023], Mamba [Gu and Dao, 2023], LinOSS, and D-LinOSS [Hasani et al., 2024, Boyer et al., 2025] are strong q -only sequence predictors. They are appropriate forecasting baselines, but they do not expose energies, port power, damping spectra, or passivity certificates. Our main quantitative suite compares against these sequence models under a shared next-step training protocol; structured and template-known comparators are reported only in stress tests where their stronger information contracts can be stated explicitly.

Table 1 summarizes the distinction that guides the method: PHAST is not only another q -only predictor, but a predictor with explicit dissipative channels and a passivity certificate.

Table 1: **Landscape of q -only dynamics predictors.** PHAST is designed for dissipative forecasting where a reader may also care about the recovered physics. “Passivity” means a by-construction storage inequality, not an empirical energy penalty.

Model family	q -only	Dissipation	Passivity	Physical channels
HNN/LNN	partial	—	yes	energy only
Dissipative SymODEN	partial	PSD	yes	energy, damping, ports
DHNN	partial	yes	partial	damping not always identifiable
Neural ODE / Latent ODE	yes	implicit	no	latent state only
SINDy / DMDc / Koopman	partial	model-dependent	no	identified coordinates
SympNets/VPT	yes	no	no	conservative geometry
GRU/LSTM/Transformer	yes	implicit	no	none
S5/LinOSS/D-LinOSS	yes	implicit/linear	no	sequence modes
PHAST	yes	PSD by construction	yes	V, M, D , ports

This comparison fixes the design target for the next section. PHAST must be causal and q -only like modern sequence models, but it must also expose the physical channels that HNN-style models make interpretable and the dissipative accounting that conservative maps omit. We therefore build the architecture around a single latent transition operator rather than separate task-specific simulators.

3 Method

PHAST builds the transition in Eq. (1) from three parts: an observer/canonicalizer, a port-Hamiltonian core, and a structure-preserving integrator (Fig. 1).

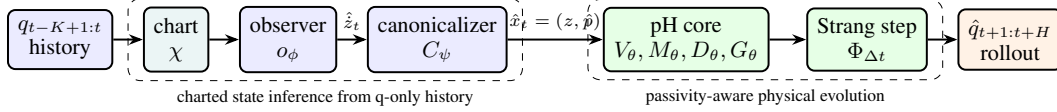


Figure 1: **PHAST pipeline.** A domain chart first represents the observed configuration on the appropriate coordinate interface; a causal observer and canonicalizer then infer latent phase state from position-only history, and the pH core evolves that state with explicit storage, geometry, PSD dissipation, and optional input/output ports.

119 3.1 Port-Hamiltonian core

120 PHAST searches over port-Hamiltonian models [Van Der Schaft and Jeltsema, 2014]

$$\begin{aligned}
 \dot{x} &= (J - R_\theta) \nabla H_\theta(x) + Gu, \\
 H_\theta(q, p) &= V_\theta(q) + \frac{1}{2} p^\top M_\theta(q)^{-1} p, \quad M_\theta(q) \succ 0, \\
 R_\theta(q) &= \begin{pmatrix} 0 & 0 \\ 0 & D_\theta(q) \end{pmatrix}, \quad D_\theta(q) = D_\theta(q)^\top \succeq 0.
 \end{aligned} \tag{2}$$

121 Here θ denotes the trainable parameter collection for the typed physical components, e.g., the potential
 122 V_θ , mass geometry M_θ , and damping channel D_θ ; components may be fixed, templated, bounded, or
 123 fully learned depending on the regime in Table 2. Geometrically, the skew matrix $J = -J^\top$ is the
 124 conservative exchange bracket, the PSD matrix R_θ is the metric channel that accounts for irreversible
 125 loss, and the port Gu is the work interface through which external actuation enters. For the canonical
 126 mechanical block, Eq. (2) is equivalently

$$\dot{q} = \partial_p H(q, p), \quad \dot{p} = -\partial_q H(q, p) - D_\theta(q) \partial_p H(q, p) + G_p u. \tag{3}$$

127 Thus V_θ determines the force landscape, M_θ determines how momentum converts to velocity, D_θ
 128 determines which velocity directions lose energy, and the port matrix G_p selects where external work
 129 enters. In the experiments, $D_\theta(q)$ multiplies velocity, so it represents viscous or state-dependent
 130 linear drag; genuinely nonlinear velocity-dependent losses can be represented by extending the
 131 damping channel to $D_\theta(q, p)$ or by adding typed dissipative ports, but that extension is not evaluated
 132 here. This is the concrete object learned from q-only histories: the observer supplies a latent (q, p) ,
 133 while the pH block explains its future through force, inertia, loss, and input channels. The measured
 134 positions are an observation channel, not necessarily the power-conjugate port output; the latter is
 135 used internally to account for work injected through G . With port output $y^{\text{port}} = G^\top \nabla H(x)$, the
 136 power balance for this architecture is

$$\dot{H}(x) = -\nabla H(x)^\top R_\theta(q) \nabla H(x) + u^\top y^{\text{port}}. \tag{4}$$

137 All forecasting experiments are autonomous ($u = 0$), so Eq. (4) certifies non-increasing learned
 138 storage in continuous time. The same equation also defines the feedback interface: a controller writes
 139 through the same work port G and reads the conjugate output y^{port} , without changing the learned
 140 plant model. Appendix F derives this port interconnection and uses a pendulum probe to show the
 141 q-only bottleneck: closed-loop quality depends on how accurately the observer recovers y^{port} from
 142 history.

143 **Proposition 1** (Learned passivity certificate). *Assume $J = -J^\top$, $R_\theta(q) \succeq 0$, and H_θ is differentiable.*
 144 *Then every smooth trajectory of Eq. (2) satisfies the power balance in Eq. (4). In particular, for*
 145 *autonomous rollouts ($u = 0$), $H_\theta(x(t_2)) \leq H_\theta(x(t_1))$ for $t_2 \geq t_1$. If the damping cap in Eq. (8)*
 146 *holds, the instantaneous dissipative power also obeys*

$$0 \leq \nabla H^\top R_\theta \nabla H = (\partial_p H)^\top D_\theta (\partial_p H) \leq (d_0 + \bar{\beta}) \|\partial_p H\|^2.$$

147 The proposition is the architectural certificate used throughout the paper. It is not an auxiliary
 148 regularizer and does not depend on fitting an energy curve after training. The model can still be
 149 wrong, but it is wrong inside a constrained class: autonomous forecasts cannot create energy through
 150 the dissipative channel, and bounded damping cannot hide arbitrary prediction error by introducing
 151 unbounded friction.

Knowledge regimes. The same pH template supports three regimes. In **KNOWN**, physical expressions for V , M , or D are fixed. In **PARTIAL**, a known template is augmented with bounded learned residuals. In **UNKNOWN**, the components are learned from data under definiteness constraints. The regimes are component-wise: the user can supply only the pieces that are trusted, and PHAST learns the remaining storage, geometry, dissipation, or port terms under the same rollout code path.

Charts as the first grey-box choice. The first modeling decision in PHAST is not the potential V , mass M , or damping D , but the coordinate chart in which the observation history is represented. If q lives on a constrained space $\mathcal{Q}$, using raw Euclidean coordinates makes the constraint a learned behavior rather than a structural fact. PHAST therefore allows a chart $z = \chi(q)$ before phase-state inference, so that positivity, periodicity, or definiteness can be handled at the interface while the pH core learns the storage and dissipation in those coordinates. The predator–prey experiment isolates this effect: replacing raw populations by log-positive coordinates removes negative-population rollouts and improves $H=100$ error under observation noise, without changing the rollout architecture. Appendix C gives the compact chart table and notes the distinction between charted observations and full Riemannian/Lie-group integrators.

Table 2: **Typed knowledge-injection interface.** The same PHAST code path is used across domains; regimes differ only in which pH components are fixed, bounded, templated, or fully learned.

Regime	$V(q)$: storage	$M(q)$: geometry	$D(q)$: dissipation	Example use
KNOWN	analytic	analytic or fixed	known law or bounded scalar	pendulum, RLC, heat
PARTIAL	template plus residual, or analytic anchor	template or low-rank learned correction	PSD low-rank with calibrated cap	robot arm friction, windy pendulum, gravity
UNKNOWN	neural	SPD neural/diagonal	PSD neural, optionally bounded	predator–prey, black-box dynamics

Concrete instantiations. The regimes in Table 2 are operational choices, not separate architectures. For a damped pendulum, $V(q)$ may be analytic while $D(q)$ is known or bounded; the model then tests whether q -only history is sufficient to infer the latent momentum. For a windy pendulum or robot arm, the conservative template is available but the dissipative channel is only partially known, so PHAST learns a PSD residual with a calibrated spectral cap. For predator–prey or other dissipative non-Hamiltonian dynamics, the storage and dissipation are learned neural components, but the update still satisfies the same storage inequality in Eq. (4).

3.2 q -only observer and canonicalizer

Since p_t is latent, PHAST first estimates a velocity-like state from history. The observer is written as a chart, a causal smoother, and a small neural residual:

$$q_{t-K+1:t} \xrightarrow{\chi} z_{t-K+1:t} \xrightarrow{S_\eta} \tilde{v}_t, \quad \hat{z}_t = \tilde{v}_t + r_\phi(z_{t-K+1:t}, \tilde{v}_{t-K+1:t}, \sigma_q) \equiv o_\phi(q_{t-K+1:t}). \quad (5)$$

In the clean strict q -only forecasting benchmarks, S_η is a finite difference and r_ϕ is a causal TCN correction. Under noisy sensing or controller-facing port estimation, finite differences amplify measurement noise as $O(\sigma_q/\Delta t)$, so PHAST uses a MAP-first residual observer:

$$\hat{z}_t = \hat{z}_t^{\text{MAP}} + r_\phi(z_{t-K+1:t}, \hat{z}_{t-K+1:t}^{\text{MAP}}, \sigma_q), \quad (6)$$

where $\hat{z}^{\text{MAP}}$ is obtained from a fixed-lag smooth trajectory fit to the noisy charted observation window. The canonicalizer maps $(z_t, \hat{z}_t)$ to phase state. In the identity chart with a known mass model this reduces to $\hat{p}_t = M(q_t)\hat{q}_t$; in nontrivial charts, phase and port targets are lifted through the chart differential or cotangent lift before supervision. When the mass model is unknown, PHAST treats $\hat{p}_t$ as a velocity-like latent sufficient for Markov rollout. This is the point at which PHAST differs from simply feeding finite differences to a recurrent model: the inferred state is immediately constrained to evolve through the pH transition in Eq. (2), so state inference is trained against long-horizon physical consistency rather than only next-frame interpolation. The observer therefore plays a role closer to proprioception than smoothing. From a short history, it must infer the hidden phase information that makes the future Markovian: direction of travel, effective momentum, unresolved damping, and local forcing. The pH core then decides which part of that inferred state should be stored as

energy, exchanged through the skew structure, or lost through PSD dissipation. This separation is important experimentally because a strong temporal model can extrapolate the observed curve while still failing to recover the physical channel that generated it. When simulator phase variables are available during training, the same observer can be given typed auxiliary targets, e.g., $\mathcal{L}_p = \|\hat{p} - p^*\|^2$, $\mathcal{L}_{\text{flow}} = \|\hat{q} - M(q)^{-1}p^*\|^2$, and $\mathcal{L}_y = \|\hat{y} - G^\top \nabla H(q, p^*)\|^2$. This is not strict q-only supervision; it is q-only deployment with phase-supervised observer training. Appendix B uses this setting to test whether PHAST’s typed observer interface improves long-horizon rollout.

3.3 Definite parameterizations and bounded damping

To enforce definiteness with fast inference, PHAST uses low-rank additive expansions. For damping in an n -dimensional momentum space, PHAST uses a nonnegative isotropic floor d_0 plus r learned rank-one dissipative directions:

$$D(q) = d_0 I + \sum_{i=1}^r \beta_i(q) k_i(q) k_i(q)^\top, \quad d_0 \geq 0, \quad \beta_i(q) \geq 0, \quad \|k_i(q)\| = 1. \quad (7)$$

Here each unit vector $k_i(q) \in \mathbb{R}^n$ is a dissipative direction and each scalar $\beta_i(q)$ is the learned loss strength along that direction. This guarantees $D(q) \succeq 0$ and yields $O(nr)$ matrix-vector products without forming a dense $n \times n$ matrix. When identifiability matters, we bound the total learned damping strength by a user-chosen cap $\bar{\beta}$:

$$\sum_i \beta_i(q) \leq \bar{\beta} \quad \Rightarrow \quad \lambda_{\max}(D(q)) \leq d_0 + \bar{\beta}. \quad (8)$$

Here $\lambda_{\max}(D(q))$ denotes the largest eigenvalue of the PSD damping matrix. This prevents D from absorbing errors in the potential, mass geometry, or inferred phase state as arbitrary friction; the damping block becomes an attribution channel rather than a generic numerical stabilizer. The rank r is therefore not just a compression knob: it specifies how many dissipative directions the model may use. Operationally, this gives a controlled path between black-box damping and hand-coded friction. If a domain expert knows a scalar drag law, PHAST can fix it. If the expert only knows that losses are low-rank or bounded, PHAST can learn the directions while preserving PSD structure. If no physical law is available, the model can still learn a dissipative operator, but the recovery metrics in Sec. 4 then become essential because the inverse problem is less identifiable. For systems with a trusted modal hierarchy, Appendix H gives an optional ordered partial-flag coordinate for the same PSD operator: $D(q) = d_0 I + F(q) \text{diag}(\alpha(q)) F(q)^\top$. In this form, $F(q) = [f_1(q), \dots, f_r(q)] \in \text{St}(n, r)$ is an orthonormal r -frame of modal directions and $\alpha_1(q) \geq \dots \geq \alpha_r(q) \geq 0$ are ordered damping strengths. This is a diagnostic and initialization interface for spectra and invariant subspaces, not a separate passivity assumption and not a claim that arbitrary q-only trajectories uniquely identify every damping hierarchy.

3.4 Integration and training

We integrate Eq. (2) with Strang splitting [Strang, 1968, Hairer et al., 2006],

$$\Phi_{\Delta t} = \Phi_D^{\Delta t/2} \circ \Phi_H^{\Delta t} \circ \Phi_D^{\Delta t/2}, \quad (9)$$

where Φ_H is a conservative symplectic update and Φ_D is a damping half-step. The same learned pH components $(V_\theta, M_\theta, D_\theta)$, and G_θ when inputs are present, can be evaluated with L internal substeps, reducing discretization error without adding physics parameters. Models are trained as causal next-step predictors from q-only histories. For a minibatch of B trajectories $\{q_{0:T-1}^{(b)}\}_{b=1}^B$, context length K , and valid teacher-forced indices

$$\mathcal{I} = \{(b, t) : b = 1, \dots, B, t = K - 1, \dots, T - 2\}, \quad N_{\mathcal{I}} = |\mathcal{I}|,$$

define

$$\hat{x}_t = C_\psi(\chi(q_t), o_\phi(q_{t-K+1:t})), \quad \hat{x}_{t+1} = \Phi_{\Delta t}(\hat{x}_t), \quad \hat{q}_{t+1} = \chi^{-1}(\Pi_z \hat{x}_{t+1}),$$

where Π_z projects phase state to charted configuration coordinates. The headline forecasting tables use the teacher-forced observation loss

$$\mathcal{L}_q = \frac{1}{N_{\mathcal{I}}} \sum_{(b,t) \in \mathcal{I}} \|\hat{q}_{t+1}^{(b)} - q_{t+1}^{(b)}\|_{\mathcal{M}}^2, \quad (10)$$

where $\|\cdot\|_{\mathcal{M}}$ uses wrapped angular error on toroidal coordinates and Euclidean error otherwise. The full objective is

$$\mathcal{L} = \mathcal{L}_q + \lambda_p \mathcal{L}_p + \lambda_{\text{flow}} \mathcal{L}_{\text{flow}} + \lambda_{\text{port}} \mathcal{L}_{\text{port}} + \lambda_{\text{pass}} \mathcal{L}_{\text{pass}} + \lambda_{\text{energy}} \mathcal{L}_{\text{energy}} + \lambda_{\text{roll}} \mathcal{L}_{\text{roll}}. \quad (11)$$

The added terms are optional typed supervision or accounting terms: momentum, chart-space flow, port output, passivity violation, energy-budget residual, and open-loop rollout loss. Appendix D gives the expanded formulas and the chart/cotangent lifts used when physical targets are supplied in non-chart coordinates. All coefficients except the teacher-forced $\mathcal{L}_q$ term are zero for the headline open-loop tables unless an ablation explicitly states otherwise. This choice keeps the supervised signal aligned with the sequence baselines: the main quantitative rows ask whether a structured transition and the declared typed priors improve forecasting when trained with the same next-step objective. When ground-truth physical channels are available, they are used for evaluation rather than as hidden supervision unless the row is explicitly marked as KNOWN.

The implementation follows this order directly: chart the observations, infer a phase state from history, advance it with the split pH transition, decode the next observation, and accumulate the selected q-only or auxiliary losses. Appendix E gives the step primitive, low-rank velocity solve, and training-objective assembly as Algorithms 1–3, with each algorithmic block cross-referenced to the corresponding term above.

4 Experiments

We evaluate PHAST on thirteen q-only benchmarks. Eight are mechanical rollouts at horizon $H = 100$: conservative/damped/windy pendulum, windy cart-pole, conservative/damped oscillator, and conservative/damped double pendulum. Five are cross-domain next-step benchmarks: RLC circuit, Lennard–Jones 3-particle cluster, heat exchange, 3-body gravity, and predator–prey. All methods receive the same q-only histories and are trained for 50 epochs with AdamW and validation-MSE model selection. Baselines are GRU, LSTM, S5, LinOSS, D-LinOSS, Transformer, and VPT, all trained as causal sequence predictors. Structured Hamiltonian, port-Hamiltonian, latent-ODE, and system-identification baselines are treated separately in stress-test ablations because they differ in what they assume: some receive finite-difference velocities, known templates, online observations, or explicit control structure rather than the same pure q-history forecasting contract.

Comparison protocol and scope. All methods receive the same position samples; PHAST forms finite differences deterministically from those samples and does not use additional sensors. The KNOWN and PARTIAL rows intentionally test the typed knowledge interface, so they combine architectural bias with supplied physical structure. The UNKNOWN rows and ablations test the data-only setting. We use a fixed 50-epoch budget to compare parameter efficiency under a shared training protocol; this does not prove that every larger baseline has reached its asymptotic optimum.

Evaluation questions. The experiments are organized around three questions. First, does the pH transition improve long-horizon q-only forecasting over strong sequence models? Second, does the typed knowledge interface work beyond a single mechanical toy problem? Third, when the ground-truth dissipative law is known, does forecast accuracy correspond to physical recovery, or can a model be accurate for the wrong reason? The third question is essential: PHAST is not only claiming lower MSE, but a usable physical interface whose internal channels can be inspected.

Metrics. For forecasting, we report one-step MSE and open-loop rollout MSE at horizon $H = 100$ where applicable. For physical recovery, when a model exposes $D(q)$ and the simulator damping law is known, we report damping R^2 and damping MAE; Appendix A also defines an energy-budget residual used for diagnostic inspection. The main point is that forecasting and identification are different axes: a model can forecast well by using damping as a stabilizer while recovering the wrong physical mechanism. Figure 2 gives the qualitative rollout story; Tables 3 and 4 give the quantitative suite evidence; Table 5 and Figure 1 show why forecast accuracy alone is not enough.

Table 3 gives the q-only evidence in compressed form. Across all thirteen benchmarks, the best PHAST regime is the best row-wise model. The gains are not only relative to weak recurrent baselines: the strongest competitors by row include Transformer, S5, LinOSS, and D-LinOSS. The best PHAST variants use 3.4k–13.7k trainable parameters, compared with roughly 16.8k–100.8k for

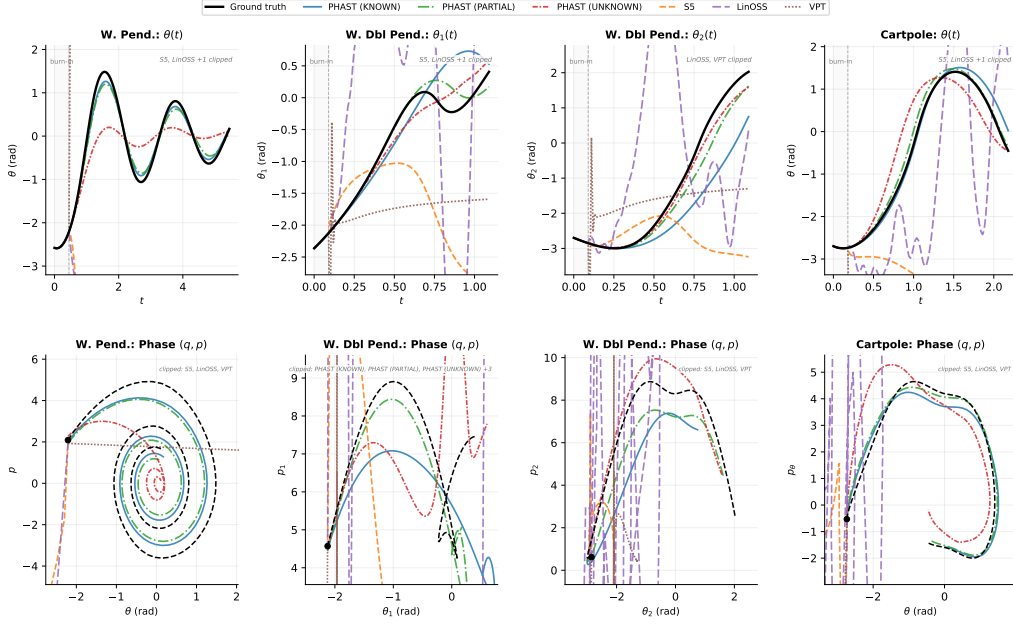


Figure 2: **Open-loop q-only rollouts and phase portraits.** PHAST tracks dissipative trajectories longer than sequence baselines on Windy Pendulum, Windy Double Pendulum, and Windy Cart-Pole. Momentum is latent; PHAST reconstructs it through the observer/canonicalizer, while baselines rely on finite-difference diagnostics.

Table 3: **Representative rows plus suite summary.** Representative mechanics rows report $H=100$ rollout MSE over 5 seeds. Summary rows aggregate the per-environment results expanded in Table 4. Lower is better; gains compare the best PHAST regime against the best non-PHAST baseline in the same row.

Setting	Best PHAST	Best baseline	Evidence
<i>Representative dissipative mechanics</i>			
Windy Pendulum	0.059 ± 0.007	0.435 ± 0.239 (D-LinOSS)	$7.4 \times$
Windy Cart-Pole	0.063 ± 0.019	0.431 ± 0.077 (S5)	$6.8 \times$
Damped double pendulum	0.320 ± 0.032	0.630 ± 0.031 (S5)	$2.0 \times$
Damped oscillator	1.1×10^{-3}	0.926 (Transformer)	8.4×10^2
<i>Cross-domain suite summary</i>			
Mechanical suite	8/8 wins	best per row	$1.5 \times -1.1 \times 10^3$
Cross-domain suite	5/5 wins	best per row	$9.0 \times -4.5 \times 10^5$
All q-only benchmarks	13/13 wins	sequence baselines	six domains

these baselines. Table 4 expands this claim row-by-row, separating mechanical rollout tasks from cross-domain next-step tasks so the reader can verify that the gains are not concentrated in a single domain. The representative mechanical rows in Table 3 are deliberately dissipative: they are the settings where conservative structure alone is insufficient and where black-box sequence models can appear stable while losing the correct phase-space geometry. The suite rows show that the same mechanism is not overfit to one simulator family; it transfers across electrical, molecular, thermal, gravitational, and ecological benchmarks by changing only the typed components supplied to the pH core.

What the gains mean. The largest numerical gains occur in systems where a small error in phase or damping compounds quickly: damped oscillators, heat exchange, Lennard–Jones interactions, and 3-body gravity. For mechanical rollout rows, black-box sequence models can fit local trends but drift when rolled forward because the learned update is not constrained to respect storage, exchange, and loss. For the cross-domain next-step rows, Table 4 tests the local-prediction contract: typed storage and dissipation improve one-step accuracy in domains where long rollout metrics require domain-specific state constraints. Smaller gains on conservative and double-pendulum settings are

296 expected: these are closer to what symplectic or generic state-space baselines handle well. PHAST’s
 297 advantage is strongest when the task depends on hidden physical accounting rather than short-term
 298 curve continuation.

299 **Reactive state estimation under a changed regime.** Appendix I stress-tests the deployed observer
 300 under post-burn-in damping shifts while keeping all learned weights fixed. On Windy Pendulum,
 301 re-running only the observer/canonicalizer on the latest q-history reduces post-shift MSE from 0.2302
 302 to 1.30×10^{-5} . On mixed-coordinate Windy Cart-Pole, the same contract remains about 3.0×10^2
 303 lower than charted receding-window MLP/TCN baselines that also re-encode the latest q-history;
 304 template-known estimators remain strong because they receive the plant form.

Table 4: **Full suite summary across thirteen q-only benchmarks.** Mechanical rows report rollout MSE at $H=100$; cross-domain rows report next-step MSE. Means use 5 seeds where available; gains are row-wise ratios.

Benchmark	Best PHAST	Best baseline	Gain
<i>Mechanical systems: rollout MSE at $H=100$</i>			
Pendulum (conservative)	0.680 ± 0.043 (PARTIAL)	2.320 ± 0.224 (Transformer)	$3.4 \times$
Pendulum (damped)	0.017 ± 0.005 (KNOWN)	0.450 ± 0.241 (D-LinOSS)	$26.5 \times$
Pendulum (windy)	0.059 ± 0.007 (PARTIAL)	0.435 ± 0.239 (D-LinOSS)	$7.4 \times$
Cart-Pole (windy)	0.063 ± 0.019 (KNOWN)	0.431 ± 0.077 (S5)	$6.8 \times$
Oscillator (conservative)	0.0010 ± 0.0002 (KNOWN; PARTIAL tie)	1.087 ± 0.299 (Transformer)	1.1×10^3
Oscillator (damped)	0.0011 ± 0.0003 (PARTIAL)	0.926 ± 0.254 (Transformer)	8.4×10^2
Double pendulum (conservative)	0.402 ± 0.047 (PARTIAL)	0.618 ± 0.028 (S5)	$1.5 \times$
Double pendulum (damped)	0.320 ± 0.032 (PARTIAL)	0.630 ± 0.031 (S5)	$2.0 \times$
<i>Cross-domain systems: next-step MSE</i>			
RLC circuit	2.63×10^{-5} (UNKNOWN)	4.81×10^{-4} (Transformer)	$18 \times$
LJ-3 cluster	4.59×10^{-10} (PARTIAL)	2.05×10^{-4} (S5)	4.5×10^5
Heat exchange	2.42×10^{-6} (KNOWN)	4.46×10^{-4} (LinOSS)	1.8×10^2
3-body gravity	4.27×10^{-8} (PARTIAL)	1.83×10^{-3} (Transformer)	4.3×10^4
Predator-prey	0.0199 (UNKNOWN)	0.179 (Transformer)	$9.0 \times$

305 **Forecasting is not identification.** Table 5 isolates why rollout error alone is insufficient. Without
 306 the damping bound in Eq. (8), PHAST-PARTIAL obtains the lowest trajectory error but assigns
 307 missing physics to an implausible damping field ($R_D^2 \ll 0$). With the bound, rollout error increases
 308 slightly while damping recovery becomes meaningful. Consistent with Proposition 1, damping
 309 should account for irreversible loss rather than act as a numerical sink for observer, potential, mass,
 310 or forcing errors. We therefore report trajectory error and physical recovery separately; Appendix
 311 Fig. 1 visualizes the corresponding damping channels and energy budgets.

Table 5: **Damping bounds turn stable rollouts into interpretable recovery.** Windy Pendulum q-only results over 5 seeds. Lower WrapMSE/MAE and higher R_D^2 are better.

Model	WrapMSE(100) ↓	R_D^2 ↑	MAE _D ↓	Interpretation
PHAST-PARTIAL, unbounded	0.059 ± 0.007	-90.15 ± 6.14	1.326 ± 0.044	accurate but nonphysical
PHAST-PARTIAL, bounded	0.071 ± 0.004	0.703 ± 0.072	0.061 ± 0.010	identifiable damping
PHAST-KNOWN	0.106 ± 0.020	0.996 ± 0.004	0.007 ± 0.004	nearly exact recovery

312 5 Limitations and conclusion

313 PHAST resolves q-only forecasting within a constrained physical class, but identifiability still depends
 314 on which physical anchors are supplied: if potential, mass, and damping are all unconstrained,
 315 damping can compensate for misspecification. Bounded damping makes that compensation visible,
 316 while modal or ordered recovery is strongest when the user supplies trusted spectral or PSD anchors.
 317 Auxiliary momentum or flow targets strengthen observer training even when deployment remains q-
 318 only, and the port interface gives a concrete path from forecasting to passivity-based feedback. Across
 319 thirteen benchmarks, this structured state-space view improves forecasting, parameter efficiency, and
 320 physical accounting over strong sequence baselines when only position or proprioceptive histories
 321 are available.

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A Metric definitions

This appendix fixes the metrics used in Sec. 4. For coordinates that live on a circle, errors are wrapped before squaring:

$$\text{wrap}(\hat{q} - q) = \text{atan2}(\sin(\hat{q} - q), \cos(\hat{q} - q)). \quad (12)$$

For mixed Euclidean–angular systems, $\|\cdot\|_{\mathcal{M}}$ denotes the norm that applies wrapped error on angular coordinates and Euclidean error on the remaining coordinates. The horizon- H open-loop rollout metric is

$$\text{MSE}_H = \frac{1}{BH} \sum_{b=1}^B \sum_{h=1}^H \|\hat{q}_{b,t+h} - q_{b,t+h}\|_{\mathcal{M}}^2, \quad (13)$$

where $\hat{q}_{b,t+h}$ is produced autoregressively from the burn-in history and no ground-truth future state is fed back during rollout.

When the simulator exposes a ground-truth damping matrix or damping channel, we compare the recovered dissipative law on an evaluation grid $\mathcal{Q}$. For matrix-valued damping, the mean absolute error and coefficient of determination are

$$\text{MAE}_D = \frac{1}{|\mathcal{Q}|n^2} \sum_{q \in \mathcal{Q}} \left\| \hat{D}(q) - D^*(q) \right\|_{1,1}, \quad (14)$$

$$R_D^2 = 1 - \frac{\sum_{q \in \mathcal{Q}} \left\| \hat{D}(q) - D^*(q) \right\|_F^2}{\sum_{q \in \mathcal{Q}} \left\| D^*(q) - \overline{D^*} \right\|_F^2}. \quad (15)$$

For scalar or vector damping laws, the same definitions are applied to the corresponding channel vector instead of the full matrix. Here $\|\cdot\|_{1,1}$ denotes the entry-wise matrix 1-norm, D^* is the

405 simulator law, and $\overline{D}^*$ is its empirical mean over $\mathcal{Q}$. If the denominator in R_D^2 is zero because the
 406 ground-truth damping law is constant on $\mathcal{Q}$, R_D^2 is undefined and we report MAE_D instead.

407 For modal damping benchmarks, we also compare the ordered invariant subspaces of the learned and
 408 true damping operators. Let $\hat{F} = [\hat{f}_1, \dots, \hat{f}_r]$ and $F^* = [f_1^*, \dots, f_r^*]$ be orthonormal eigenframes
 409 sorted by descending damping strength, and let $\alpha_1^* \geq \dots \geq \alpha_r^* \geq 0$ be the true modal strengths. For
 410 each level k , define

$$\hat{S}_k = \text{span}(\hat{f}_1, \dots, \hat{f}_k), \quad S_k^* = \text{span}(f_1^*, \dots, f_k^*).$$

411 If $\sigma_i^{(k)}$ are the singular values of $[\hat{f}_1, \dots, \hat{f}_k]^\top [f_1^*, \dots, f_k^*]$, the normalized Grassmann chordal
 412 distance at level k is

$$d_k = \sqrt{\frac{1}{k} \sum_{i=1}^k \left(1 - (\sigma_i^{(k)})^2\right)} \in [0, 1]. \quad (16)$$

413 The reported flag alignment is

$$\text{Align}_{\text{FI}} = 1 - \sum_{k=1}^r w_k d_k, \quad w_k = \frac{\alpha_k^*}{\sum_{j=1}^r \alpha_j^*}, \quad (17)$$

414 with uniform weights if all true strengths vanish. This metric is invariant to signs and basis rotations
 415 inside a level- k subspace. It reduces to ordinary eigenvector alignment only when all relevant spectral
 416 gaps are well separated. We use this chordal flag metric as a cheap representative of the natural
 417 quotient geometry: near the diagonal it agrees to first order with the Riemannian metric induced from
 418 the Stiefel cover, while avoiding explicit geodesic computations.

419 The energy-budget residual checks whether predicted storage decay is explained by the learned
 420 dissipative power. For autonomous systems,

$$\varepsilon_E(t) = \frac{H(\hat{x}_{t+1}) - H(\hat{x}_t)}{\Delta t} + \nabla H(\hat{x}_t)^\top R(\hat{q}_t) \nabla H(\hat{x}_t). \quad (18)$$

421 Small residual means the model’s change in storage is consistent with its own dissipative channel.
 422 Large residual means the learned rollout is using numerical or modeling error rather than the pH
 423 power balance to explain decay.

424 B Phase-supervised q-only observer probe

425 The main benchmark uses strict q-only supervision: position histories and next-position targets. Here
 426 we test a stronger but operationally common regime. The deployed model still receives only $q_{t-K+1:t}$,
 427 but simulator momenta are available during training as auxiliary observer targets. For systems with
 428 configuration-dependent mass,

$$p_t^* = M(q_t) \dot{q}_t, \quad \dot{q}_t^* = M(q_t)^{-1} p_t^*,$$

429 so supervising momentum and supervising flow are different physical signals. We add one typed
 430 observer loss at a time:

$$\mathcal{L}_p = \|\hat{p}_t - p_t^*\|^2, \quad \mathcal{L}_{\dot{q}} = \|\hat{\dot{q}}_t - \dot{q}_t^*\|^2, \quad \mathcal{L}_y = \|\hat{y}_t^{\text{port}} - \dot{q}_t^*\|^2.$$

431 The effect is not captured by one-step q-MSE: typed observer losses slightly worsen local interpolation
 432 but can improve $H=50$ rollout. The best target is environment-dependent, so the useful supervision
 433 signal is a property of the physical coordinates rather than a universal momentum recipe. The
 434 experiment shows that PHAST exposes typed observer targets: simulator-only phase supervision can
 435 improve the latent variables used by the pH rollout without changing q-only deployment.

436 C Domain charts

437 The first grey-box choice in PHAST is the coordinate chart in which the observation history is
 438 represented. For an observed state $q \in \mathcal{Q}$, a chart $z = \chi(q)$ keeps observations and decoded
 439 predictions on the correct target space before the observer infers latent phase variables. This is

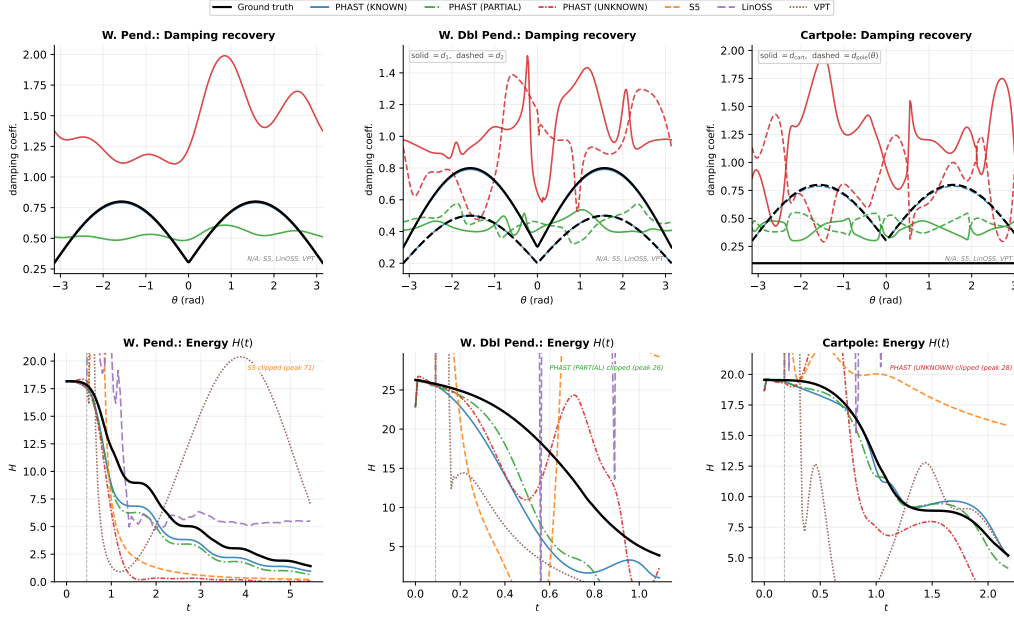


Figure 1: **Physical recovery diagnostics.** PHAST exposes damping channels and energy-budget diagnostics that explain forecasted decay. The gallery separates a single angular damping profile, two joint-wise wind profiles under coupled dynamics, and mixed cart friction plus angular wind damping.

Table 1: **Phase-supervised q-only observer probe.** Mean $\pm$ std over 3 seeds at observation noise $\sigma = 0.03$. Auxiliary phase targets are used only during training; deployment remains q-only.

Environment	Observer loss	q MSE $\downarrow$	$H=50$ rollout $\downarrow$	Change
Windy Cart-Pole	strict q-only base	0.0026 ± 0.0001	0.4037 ± 0.0795	–
Windy Cart-Pole	flow $\dot{q}$	0.0028 ± 0.0001	0.3638 ± 0.0457	–9.9%
Windy Cart-Pole	momentum p	0.0031 ± 0.0001	0.4045 ± 0.2030	+0.2%
Windy Cart-Pole	flow + p + port output	0.0028 ± 0.0001	0.3582 ± 0.0214	–11.3%
Damped Double Pendulum	strict q-only base	0.0026 ± 0.0001	0.2893 ± 0.0821	–
Damped Double Pendulum	flow $\dot{q}$	0.0028 ± 0.0001	0.2993 ± 0.0898	+3.5%
Damped Double Pendulum	momentum p	0.0029 ± 0.0001	0.2410 ± 0.0587	–16.7%
Damped Double Pendulum	flow + p + port output	0.0028 ± 0.0001	0.2791 ± 0.0688	–3.5%

Table 2: **Coordinate knowledge as a PHAST interface.** Charts encode hard state-space constraints before latent phase inference; richer Riemannian or Lie-group backends would additionally use retractions, metrics, or noncanonical brackets.

State type	Target manifold	Chart/interface	Use in PHAST
Positive variables	$\mathbb{R}_{>0}^n$	log chart or positive decoder	preserves positivity for populations, concentrations, rates, or conductances
Angles	$\mathbb{T}^n$	wrapped loss, unwrap, or Fourier features	avoids branch-cut discontinuities
Rotations	$SO(3)$ or quaternions	Lie algebra chart, quaternion, or retraction	keeps orientations on the rotation manifold
SPD quantities	$SPD(n)$	Cholesky, log-Cholesky, or matrix log	preserves definiteness of mass, covariance, or metric objects
Product systems	$\mathcal{Q}_1 \times \dots \times \mathcal{Q}_m$	per-factor chart and product loss	supports mixed Euclidean, angular, articulated, molecular, or contact coordinates

separate from component knowledge in V , M , D , and ports: the chart encodes the state-space constraint, while the pH components define how storage, geometry, dissipation, and work are modeled in those coordinates.

The predator–prey sweep in Table 3 isolates this interface without changing the rollout architecture. Raw populations let the model leave the biologically valid state space, while log-positive coordinates preserve positivity and reduce long-horizon error under observation noise.

Table 3: **Log-positive charts preserve the predator–prey state space.** PHAST is trained with the same rollout architecture under increasing observation noise. Raw populations produce negative-population rollouts; the log chart removes that failure mode and improves $H=100$ error.

Noise std.	Raw $H=100$ MSE	Log-chart $H=100$ MSE	Raw negative rate
0.00	30.51	12.33	22.16%
0.01	27.98	11.51	17.35%
0.02	27.14	13.86	16.09%
0.05	30.85	23.81	28.74%

A chart alone is not a full manifold dynamics model. Canonical momenta transform by the cotangent lift through the inverse chart: if $z = \chi(q)$ and $q = \chi^{-1}(z)$, then $p_z = D\chi^{-1}(z)^\top p_q$. For noncanonical variables, such as body-frame rigid-body coordinates, the interconnection matrix J may instead become state-dependent. The current experiments use chart/product-coordinate interfaces, but the same PHAST separation is compatible with richer geometric backends: a Lie-group or Riemannian implementation can replace the straight chart update by a retraction or exponential-map step, and can replace the canonical bracket by the appropriate Lie–Poisson or metric-compatible bracket when the application requires global manifold integration.

D Training loss ledger

Section 3.4 gives the training contract: the headline forecasting experiments optimize the teacher-forced q-only loss $\mathcal{L}_q$, while all other terms in Eq. (11) are optional and enabled only in explicitly marked ablations or regimes. This section records the expanded definitions so that auxiliary supervision cannot be confused with hidden supervision in the main tables.

Let $\mathcal{I} = \{(b, t) : b = 1, \dots, B, t = K - 1, \dots, T - 2\}$ and $N_{\mathcal{I}} = |\mathcal{I}|$ as in Sec. 3.4. For each $(b, t) \in \mathcal{I}$, PHAST charts the observation $z_t^{(b)} = \chi(q_t^{(b)})$, estimates $\hat{x}_t^{(b)} = (\hat{z}_t^{(b)}, \hat{p}_t^{(b)})$, advances it to $\hat{x}_{t+1}^{(b)} = \Phi_{\Delta t}(\hat{x}_t^{(b)})$, and decodes $\hat{q}_{t+1}^{(b)} = \chi^{-1}(\Pi_z \hat{x}_{t+1}^{(b)})$. When simulator phase targets are available, they are first mapped into chart coordinates: velocities by the chart differential and momenta by the cotangent lift described in Appendix C.

Typed observer and port targets. The optional phase/port terms are

$$\begin{aligned}
\mathcal{L}_p &= \frac{1}{N_{\mathcal{I}}} \sum_{(b,t) \in \mathcal{I}} \left\| \hat{p}_t^{(b)} - p_t^{\star(b)} \right\|^2, \\
\mathcal{L}_{\text{flow}} &= \frac{1}{N_{\mathcal{I}}} \sum_{(b,t) \in \mathcal{I}} \left\| \hat{z}_t^{(b)} - \dot{z}_t^{\star(b)} \right\|^2, \\
\mathcal{L}_{\text{port}} &= \frac{1}{N_{\mathcal{I}}} \sum_{(b,t) \in \mathcal{I}} \left\| \hat{y}_t^{\text{port}(b)} - y_t^{\star(b)} \right\|^2, \quad \hat{y}_t^{\text{port}} = G^\top \nabla H_\theta(\hat{x}_t).
\end{aligned} \tag{19}$$

Here p_t^* is a simulator or analytic momentum target, $\dot{z}_t^*$ is the corresponding chart-space flow target, and y_t^* is an optional conjugate port-output target. The deployed model still consumes only the declared observation history; these terms are training-time observer probes when such typed targets are available.

Energy and passivity accounting. For autonomous passive rollouts, any increase in stored energy is a violation of the continuous-time certificate in Proposition 1. The optional passivity penalty is

$$\mathcal{L}_{\text{pass}} = \frac{1}{N_{\mathcal{I}}} \sum_{(b,t) \in \mathcal{I}} \max\left(0, H_\theta(\hat{x}_{t+1}^{(b)}) - H_\theta(\hat{x}_t^{(b)})\right). \tag{20}$$

471 When a port input u_t is present, the one-step energy ledger is

$$\delta_H^{(b,t)} = \frac{H_\theta(\hat{x}_{t+1}^{(b)}) - H_\theta(\hat{x}_t^{(b)})}{\Delta t} + \hat{v}_t^{(b)\top} D_\theta(\hat{z}_t^{(b)}) \hat{v}_t^{(b)} - u_t^{(b)\top} \hat{y}_t^{\text{port}(b)}, \quad \hat{v}_t = M_\theta(\hat{z}_t)^{-1} \hat{p}_t, \quad (21)$$

472 and the optional energy-budget loss is

$$\mathcal{L}_{\text{energy}} = \frac{1}{N_{\mathcal{I}}} \sum_{(b,t) \in \mathcal{I}} \left| \delta_H^{(b,t)} \right|. \quad (22)$$

473 For zero input, the port-work term drops out and the ledger asks whether predicted storage decay is
474 explained by learned dissipated power.

475 **Open-loop rollout term.** Some ablations add an autoregressive rollout loss after teacher-forced
476 training. For sampled rollout starts $\mathcal{R} \subset \{(b, t_0) : t_0 = K - 1, \dots, T - 1 - H_{\text{roll}}\}$, initialize $\hat{x}_0^{(b,t_0)}$
477 from the observed history ending at t_0 and roll forward without future ground-truth states:

$$\begin{aligned} \tilde{x}_h^{(b,t_0)} &= \Phi_{\Delta t}(\tilde{x}_{h-1}^{(b,t_0)}), & \tilde{q}_h^{(b,t_0)} &= \chi^{-1}(\Pi_z \tilde{x}_h^{(b,t_0)}), \\ \mathcal{L}_{\text{roll}} &= \frac{1}{|\mathcal{R}| H_{\text{roll}}} \sum_{(b,t_0) \in \mathcal{R}} \sum_{h=1}^{H_{\text{roll}}} \left\| \tilde{q}_h^{(b,t_0)} - q_{t_0+h}^{(b)} \right\|_{\mathcal{M}}^2. \end{aligned} \quad (23)$$

478 Unless a table row explicitly enables this term, the reported forecasting comparisons are trained with
479 the same one-step q-only signal as the sequence baselines.

480 E Algorithmic details

481 This section expands the compact implementation description in Sec. 3.4. The color cues match the
482 method components: chart/interface operations, observer/canonicalizer state inference, conservative-
483 dissipative pH evolution, and loss assembly.

Algorithm 1 PHAST step $\Phi_{\Delta t}(q, p) \rightarrow (q^+, p^+)$ (Strang splitting with leapfrog core; constant mass)

Require: Phase state $(q, p) \in \mathbb{R}^{2n}$, time step Δt , energy terms V, M , damping D

- 1: **Dissipative half-step: irreversible momentum loss** Eq. (9); Eq. (8)
 - 2: $v \leftarrow M^{-1}p$
 - 3: $p \leftarrow p - \frac{\Delta t}{2} D(q)v$
 - 4: **Conservative Hamiltonian transport** Eq. (2)
 - 5: $p_{1/2} \leftarrow p - \frac{\Delta t}{2} \nabla V(q)$
 - 6: $v_{1/2} \leftarrow M^{-1}p_{1/2}$
 - 7: $q^+ \leftarrow q + \Delta t v_{1/2}$
 - 8: $p^+ \leftarrow p_{1/2} - \frac{\Delta t}{2} \nabla V(q^+)$
 - 9: **Symmetric dissipative closing half-step** Strang closure in Eq. (9)
 - 10: $v^+ \leftarrow M^{-1}p^+$
 - 11: $p^+ \leftarrow p^+ - \frac{\Delta t}{2} D(q^+)v^+$
 - 12: **return** (q^+, p^+)
-

Algorithm 2 Compute $v = M^{-1}p$ for diagonal-plus-low-rank mass

Require: $p \in \mathbb{R}^n$, diagonal $d \in \mathbb{R}_{>0}^n$ defining $\Lambda = \text{diag}(d)$, and $U \in \mathbb{R}^{n \times r}$ such that $M = \Lambda + UU^\top$

- 1: $z \leftarrow p \oslash d$ $\{z = \Lambda^{-1}p; O(n)$ element-wise division $\}$
 - 2: $Z \leftarrow U \oslash d$ $\{Z = \Lambda^{-1}U; O(nr)$ column-wise division $\}$
 - 3: $A \leftarrow I_r + U^\top Z$ $\{O(nr^2)\}$
 - 4: $c \leftarrow A^{-1}(U^\top z)$ $\{O(r^3)\}$
 - 5: $v \leftarrow z - Zc$ $\{O(nr)\}$
 - 6: **return** v
-

Algorithm 3 PHAST training objective assembly (q-only by default; optional phase and rollout terms)

Require: Physical trajectories $\{q_{0:T-1}^{(b)}\}_{b=1}^B$ on $\mathcal{Q}$; chart χ and decoder χ^{-1} ; chart-state projection Π_z ; context K ; optional phase/port targets; weights λ .

```

1:  $\mathcal{L}_q, \mathcal{L}_p, \mathcal{L}_{\text{flow}}, \mathcal{L}_{\text{port}}, \mathcal{L}_{\text{pass}}, \mathcal{L}_{\text{energy}}, \mathcal{L}_{\text{roll}} \leftarrow 0$ 
2: for  $b = 1$  to  $B$  do
3:   Chart physical observations Appendix C
4:    $z_t^{(b)} \leftarrow \chi(q_t^{(b)})$  for  $t = 0, \dots, T - 1$ 
5:   for  $t = K - 1$  to  $T - 2$  do
6:     Infer phase variables from charted history Eq. (5); Eq. (6)
7:      $\hat{z}_t \leftarrow S_\eta(z_{t-K+1:t}) + r_\phi(z_{t-K+1:t}, S_\eta, \sigma_q)$ 
8:      $\hat{x}_t = (\hat{z}_t, \hat{p}_t) \leftarrow C_\psi(z_t^{(b)}, \hat{z}_t)$ 
9:     if optional phase/port targets are available then
10:       Map targets into chart coordinates using the chart differential/cotangent lift.
11:       Accumulate  $\mathcal{L}_p, \mathcal{L}_{\text{flow}}, \mathcal{L}_{\text{port}}$  from Eq. (19); Appendix B
12:     end if
13:     Roll out with the port-Hamiltonian core Algorithm 1; Eq. (9)
14:      $\hat{x}_{t+1} \leftarrow \Phi_{\Delta t}(\hat{x}_t)$ 
15:      $\hat{q}_{t+1} \leftarrow \chi^{-1}(\Pi_z \hat{x}_{t+1})$ 
16:     Add teacher-forced q-only loss Eq. (10)
17:      $\mathcal{L}_q += \|\hat{q}_{t+1} - q_{t+1}^{(b)}\|_{\mathcal{M}}^2$ 
18:     if energy terms are enabled and phase information is available then
19:        $v_t \leftarrow M^{-1} \hat{p}_t$ 
20:       Add passivity and energy-budget checks Eq. (22)
21:        $\mathcal{L}_{\text{pass}} += \max(0, H(\hat{x}_{t+1}) - H(\hat{x}_t))$ 
22:        $\Delta H_t \leftarrow H(\hat{x}_{t+1}) - H(\hat{x}_t)$ 
23:        $\hat{y}_t^{\text{port}} \leftarrow G^\top \nabla H(\hat{x}_t)$ 
24:        $\delta_H \leftarrow \Delta H_t / \Delta t + v_t^\top D(\hat{z}_t) v_t - u_t^\top \hat{y}_t^{\text{port}}$ 
25:        $\mathcal{L}_{\text{energy}} += |\delta_H|$ 
26:     end if
27:   end for
28:   if open-loop rollout loss is enabled then
29:     Unroll without teacher forcing Eq. (23)
30:     Choose  $t_0 \in \{K - 1, \dots, T - 1 - H_{\text{roll}}\}$  and initialize  $\tilde{x}_0$  from  $z_{t_0-K+1:t_0}^{(b)}$ .
31:     for  $h = 1$  to  $H_{\text{roll}}$  do
32:        $\tilde{x}_h \leftarrow \Phi_{\Delta t}(\tilde{x}_{h-1}); \tilde{q}_h \leftarrow \chi^{-1}(\Pi_z \tilde{x}_h)$ 
33:        $\mathcal{L}_{\text{roll}} += \|\tilde{q}_h - q_{t_0+h}^{(b)}\|_{\mathcal{M}}^2$ 
34:     end for
35:   end if
36: end for
37: Assemble objective and update Eq. (11); Sec. 3.4
38:  $\mathcal{L}_{\text{aux}} \leftarrow \lambda_p \mathcal{L}_p + \lambda_{\text{flow}} \mathcal{L}_{\text{flow}} + \lambda_{\text{port}} \mathcal{L}_{\text{port}} + \lambda_{\text{pass}} \mathcal{L}_{\text{pass}} + \lambda_{\text{energy}} \mathcal{L}_{\text{energy}} + \lambda_{\text{roll}} \mathcal{L}_{\text{roll}}$ 
39:  $\mathcal{L} \leftarrow \mathcal{L}_q + \mathcal{L}_{\text{aux}}$ 
40: Update parameters with  $\nabla \mathcal{L}$ 

```

484 F Port-based feedback interface

485 Forecasting and passivity-based control use the same pH accounting. In the main experiments we
 486 set $u = 0$ and evaluate autonomous rollouts. Attaching feedback instead chooses u through an
 487 external controller that exchanges power through the port matrix G and reads the conjugate output
 488 $y^{\text{port}} = G^\top \nabla H$. The learned plant already exposes the control interface in Eq. (4); attaching
 489 feedback amounts to connecting another pH block through the port variables. This section makes

that interface explicit and reports a small pendulum probe that isolates the port-output estimation problem.

Plant and controller ports. A PHAST plant with actuation is

$$\dot{x}_p = (J_p - R_p)\nabla H_p(x_p) + G_p u_p, \quad y_p = G_p^\top \nabla H_p(x_p). \quad (24)$$

Here $u_p^\top y_p$ is input power. A dynamic pH controller has its own storage $H_c(\xi)$ and port pair (u_c, y_c) . The standard power-preserving interconnection is

$$u_c = y_p, \quad u_p = -y_c + v, \quad y_p^\top u_p + y_c^\top u_c = y_p^\top v. \quad (25)$$

Thus, when $v = 0$, the plant and controller exchange energy but the interconnection creates none. With exact port output and $K_d \succeq 0$, choosing $v = -K_d y_p$ injects damping. In q-only feedback we replace y_p by an estimate $\hat{y}_p$, so the closed-loop quality depends on port-output estimation error rather than on a different controller algebra.

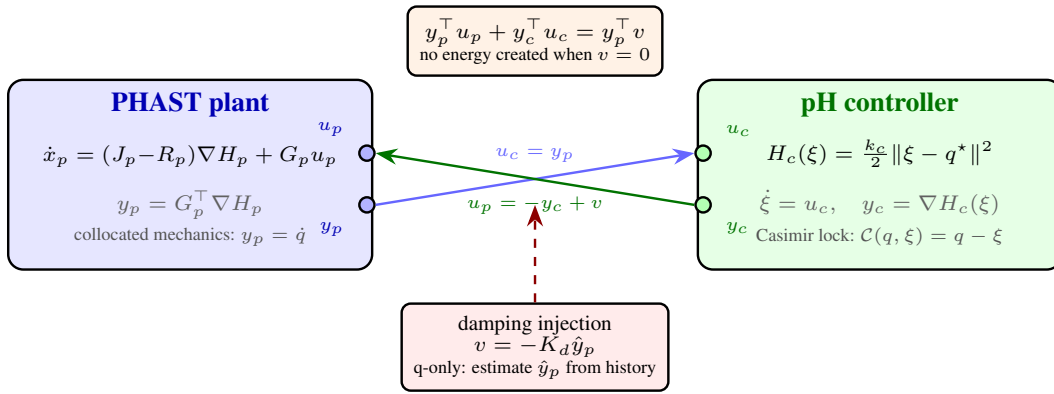


Figure 2: **Controller interface exposed by PHAST ports.** The learned plant model supplies the same input/output variables used by passivity-based control. The pH controller stores a shaped objective, the interconnection transfers power without creating energy, and damping injection depends on the estimated plant port output $\hat{y}_p$.

Casimir lock and q-only bottleneck. For the one-degree-of-freedom pendulum probe, the controller state ξ stores the desired potential around q^* , and the Casimir $\mathcal{C}(q, \xi) = q - \xi$ locks the controller coordinate to the plant configuration under the ideal interconnection. With collocated torque actuation, $y_p = \dot{q}$. Therefore the closed-loop bottleneck under q-only sensing is not the algebra of the controller; it is the online estimate $\hat{y}_p \approx \dot{q}$ used for damping injection. This makes the probe a direct test of PHAST’s observer–port interface.

Pendulum interface probe. Table 4 keeps the Energy–Casimir controller and gains fixed and varies only the port-output estimate. The oracle row verifies the controller structure. Finite differences are stable at moderate noise but inject substantially more effort because velocity estimates are noisy. MAP smoothing and learned observers reduce this error; the PHAST full-state row isolates model-based port output when (q, p) is observed.

The result should be read as port-interface evidence: PHAST exposes the plant variables needed by passivity-based feedback, and q-only control quality depends on the observer’s ability to estimate the conjugate port output. Multi-DOF tracking and controller synthesis can then be built on the same port variables, while the main benchmark focuses on the autonomous forecasting model.

G Discrete-time and damping-cap caveats

Proposition 1 in the main text is a continuous-time passivity statement. The implementation uses a finite-step Strang splitting map, so a long rollout can show small numerical energy discrepancies even

Table 4: **Port-output estimation controls closed-loop quality.** Pendulum Energy–Casimir probe at observation noise $\sigma = 0.01$. All rows use the same controller; only $\hat{y}_p$ changes.

Regime	Port estimate $\hat{y}_p$	Success $\uparrow$	Final error $\downarrow$	Effort $\downarrow$	$ \hat{y}_p - y_p \downarrow$
Oracle full-state	$\dot{q}$	1.00	5.49×10^{-3}	262.9	0.000
q-only baseline	finite difference	1.00	1.67×10^{-2}	342.2	1.117
q-only smoother	MAP estimate	1.00	1.17×10^{-2}	324.8	0.133
q-only learned observer	FD+TCN	1.00	2.39×10^{-2}	335.5	0.172
q-only PHAST observer	PHAST observer	1.00	1.93×10^{-2}	327.7	0.979
full-state PHAST plant	$\partial \hat{H} / \partial p$	1.00	5.32×10^{-3}	245.1	0.011

when the continuous vector field is passive. The energy-budget residual above is intended to expose this gap: it measures whether the observed storage change is explained by the model’s own dissipative power at the chosen timestep. In practice, reducing Δt or increasing the number of internal substeps decreases this numerical component; the main tables use the same integration protocol across PHAST variants.

Integrator substeps. PHAST can compose L evaluations of the same split pH map inside one observed sampling interval,

$$x_{t+1} = \underbrace{\Phi_{\Delta t/L} \circ \dots \circ \Phi_{\Delta t/L}}_{L \text{ substeps}}(x_t). \quad (26)$$

Substepping is a numerical accuracy knob, not extra model depth: every substep reuses the same learned storage, mass, damping, and optional port components ($V_\theta, M_\theta, D_\theta, G_\theta$). It plays the role of evaluating the learned physical vector field at a smaller internal step while preserving the Strang split between conservative transport and dissipative loss. This is different from replacing PHAST by a generic RK4 solver; RK4 is a high-order black-box time integrator, while the split map is chosen so that the discrete update follows the pH energy accounting.

Table 5: **Effect of PHAST internal substeps at coarse sampling.** Mean $\pm$ std over 5 seeds for PHAST (PARTIAL, q-only) with fixed internal step $\delta t_s = \Delta t/L$ for $s = 1, \dots, L$. Increasing L reduces discretization error at the expected compute cost while adding no learned parameters.

Environment	Metric	Δt	$L=1$	$L=4$
Windy Cart-Pole	rollout MSE ($H=100$)	0.05	0.134 ± 0.020	0.126 ± 0.016
Double Pendulum	rollout MSE ($H=100$)	0.04	2.412 ± 0.071	2.366 ± 0.088

The damping cap $\bar{\beta}$ in Eq. (8) should not be read as an oracle guarantee of true physical identifiability. It is a design contract that bounds how much instantaneous dissipative power the learned channel can introduce. Different environments instantiate this same PSD object differently. For a windy pendulum, the learned scalar channel should recover a configuration-dependent air-drag law. For Windy Cart-Pole, the diagonal entries separate floor/cart friction from angular wind damping on the pole. For articulated or molecular systems, a full PSD channel can represent modal loss: eigenvectors identify dissipative directions and eigenvalues determine how strongly velocity in those directions is attenuated. The spectral cap is therefore a physical attribution constraint: it limits the largest possible loss rate rather than merely regularizing a neural weight matrix. When physical bounds are available, $\bar{\beta}$ can be set from them. When they are not, a practical selection rule is to sweep a small set of caps and choose the Pareto point that preserves rollout accuracy while reducing energy-budget residuals and improving damping recovery on validation trajectories. This is the role of the bounded Windy Pendulum result in Table 5: the cap sacrifices a small amount of trajectory accuracy to prevent dissipation from absorbing errors that belong to other model components.

H Ordered partial-flag damping

The low-rank PSD damping form in Eq. (7) is sufficient for passivity, but its factors are not unique. Writing $K(q) = [\sqrt{\beta_1(q)}k_1(q), \dots, \sqrt{\beta_r(q)}k_r(q)]$ gives $D(q) = d_0 I + K(q)K(q)^\top$. For any orthogonal $Q \in O(r)$, $(KQ)(KQ)^\top = KK^\top$, so the raw columns of K have a continuous gauge

548 symmetry. For safety, physical recovery, and mode diagnostics, the object that is identifiable is
 549 therefore the dissipative matrix, its spectrum, and its invariant subspaces, not a named neural factor.
 550 Flag manifolds are the standard geometry of nested subspaces [Ye et al., 2022, Szwagier and Pennec,
 551 2025, Mankovich et al., 2025]. The ordered parameterization should be read as a Stiefel cover of
 552 this quotient: if an orthogonal block Q commutes with $\text{diag}(\alpha)$, then FQ and F produce exactly the
 553 same damping matrix. Losses that depend on D are therefore constant along these gauge orbits; the
 554 implementation chooses an orthonormal representative, while the metric below evaluates the quotient
 555 object. The ordered partial-flag form makes this explicit:

$$D(q) = d_0 I + F(q) \text{diag}(\alpha_1(q), \dots, \alpha_r(q)) F(q)^\top, \quad (27)$$

$$F(q) \in \text{St}(n, r), \quad \alpha_1(q) \geq \dots \geq \alpha_r(q) \geq 0. \quad (28)$$

556 Here $F(q) = [f_1(q), \dots, f_r(q)] \in \mathbb{R}^{n \times r}$ is an orthonormal frame on the Stiefel manifold, and the
 557 nested spans

$$\text{span}(f_1) \subset \text{span}(f_1, f_2) \subset \dots \subset \text{span}(f_1, \dots, f_r)$$

558 define a partial flag of dissipative subspaces. The ordered nonnegative spectrum $\alpha_1, \dots, \alpha_r$ removes
 559 permutation ambiguity and turns the arbitrary low-rank factors into ordered eigenspace coordinates.
 560 If the relevant eigenvalue gaps are nonzero, the ordered eigenspaces are determined up to sign
 561 conventions. If gaps vanish or become small, rotations within the corresponding eigenspace remain
 562 a gauge symmetry; the identifiable object coarsens from individual directions to a spectral band.
 563 Diagnostics should then be reported as subspace or flag distances rather than as a single labeled
 564 direction. This is why Table 6 reports the flag alignment from Appendix A, not only per-vector
 565 eigenvector matches.

566 **When this coordinate is useful.** Product configuration spaces and partial flags encode different
 567 structure. Articulated pose models and molecular conformers are often represented on product
 568 manifolds, e.g., products of Lie groups or torsion-angle circles [Vemulapalli et al., 2014, D unkel et al.,
 569 2024, Jing et al., 2022, Ganea et al., 2021]. Those product coordinates describe the factors of the
 570 state space. The partial flag instead orders dissipative subspaces inside the tangent or cotangent space
 571 of that product. It is therefore a useful damping coordinate when the true system has a persistent
 572 hierarchy—for example proximal-to-distal joint groups, global-to-local deformation modes, or slow-
 573 to-fast molecular modes. If the target damping is merely diagonal per coordinate with crossing or
 574 weakly separated strengths, bounded non-orthogonal damping may be the better inductive bias.

575 **Implementation.** In the ordered variant, an unconstrained network emits a matrix $A(q) \in \mathbb{R}^{n \times r}$
 576 and logits $\ell(q) \in \mathbb{R}^r$. The implementation maps $A(q)$ to $F(q)$ with a differentiable Householder
 577 reflection chain; a reduced-QR parameterization with fixed signs is an equivalent alternative. Either
 578 construction enforces $F(q)^\top F(q) = I_r$ exactly up to floating-point tolerance. The spectrum is
 579 produced by nonnegative increments, for example $\alpha_i(q) = \sum_{j=i}^r \text{softplus}(\ell_j(q))$, followed by an
 580 optional rescaling when the spectral cap in Eq. (8) is active. This makes orthonormality and monotone
 581 ordering hard constraints in the rollout, not soft penalties.

582 **PSD/modal anchor initialization.** The ordered form is most useful when the domain sup-
 583 plies a full PSD damping anchor rather than only a scalar friction estimate. Let a calibration
 584 procedure, local linear fit, or trusted domain model provide $D_0 \succeq 0$ with eigendecomposi-
 585 tion $D_0 = U \text{diag}(\lambda_1, \dots, \lambda_n) U^\top$ and $\lambda_1 \geq \dots \geq \lambda_n \geq 0$. For rank r , PHAST initializes
 586 $F = [u_1, \dots, u_r]$, sets $d_0 = \lambda_{r+1}$ when $r < n$ (or a small nonnegative floor when $r = n$), and sets
 587 $\alpha_i = \lambda_i - d_0$ for $i \leq r$. This is the full-operator analogue of scalar damping initialization: instead of
 588 averaging D_0 into one friction coefficient, it preserves the modal directions and spectral gaps that
 589 determine dissipated power. The initialization is optional and should be used only when the PSD
 590 anchor is physically meaningful.

591 The per-level columns show what the operator metric hides. The bounded non-orthogonal form
 592 forecasts reasonably but recovers neither the operator nor the dissipative geometry. The ordered
 593 coordinate improves the flag alignment but is not sufficient by itself. PSD initialization creates a
 594 different failure mode: it preserves the spectrum (5.88×10^{-4} eigenvalue MAE) while the frame can
 595 still rotate away from the physical modal directions. Adding dissipated-power matching improves
 596 both the forecast and the physically active geometry: the dominant line is recovered to $d_1 = 0.096$ and
 597 the top-three dissipative subspace to $d_3 = 0.088$. The remaining $d_2 = 0.707$ is therefore informative

Table 6: **Ordered damping with a molecular product-space anchor.** The modal-damped LJ-3 q-only benchmark has six product coordinates but a damping hierarchy in internal stretch/bend modes and weak center-of-mass modes. Lower is better except flag alignment. Align_{F1} is the spectrally weighted nested-subspace metric defined in Appendix A; d_k is the level- k Grassmann distance. The ordered coordinate alone is inspectable but not sufficient for exact recovery; a PSD/modal anchor plus dissipated-power matching improves rollout, dominant-subspace recovery, and modal-power attribution.

Damping form	Rollout	Rel. D err.	Eig. MAE	Align_{F1}	d_1	d_2	d_3	Power MSE
Bounded non-orthogonal	7.39×10^{-5}	0.789	0.205	0.243	0.965	0.757	0.599	6.50×10^{-2}
Ordered partial flag	6.32×10^{-5}	0.842	0.250	0.356	0.698	0.727	0.586	6.85×10^{-2}
Ordered + PSD init.	8.71×10^{-5}	0.652	5.88×10^{-4}	0.360	0.799	0.687	0.572	6.05×10^{-3}
Ordered + PSD init., core frozen	8.85×10^{-5}	3.65×10^{-7}	1.02×10^{-7}	1.000	0	0	0	4.89×10^{-15}
Ordered + PSD init. + power loss	2.31×10^{-5}	0.282	9.40×10^{-4}	0.743	0.096	0.707	0.088	1.32×10^{-4}

rather than hidden by the average alignment score: the model recovers the strongest mode and the cumulative three-mode dissipative subspace, but not the internal ordering of the middle modes. This is the purpose of reporting flag-level metrics separately from operator-level error and eigenvalue error. The spectral cap in Eq. (8) has a direct interpretation in this coordinate system:

$$\sum_{i=1}^r \alpha_i(q) \leq \bar{\beta} \implies \lambda_{\max}(D(q)) \leq d_0 + \bar{\beta}. \quad (29)$$

Thus the cap bounds how much instantaneous dissipative power the learned channel can introduce. This is useful in q-only inverse problems because otherwise damping may absorb errors that actually belong to the observer, potential, mass geometry, or unmodeled forcing. The ordered partial-flag view gives the PSD damping block a mathematically cleaner coordinate system while preserving the same PHAST rollout architecture.

I Dynamic regime-shift stress test

The main suite evaluates fixed train/test regimes. Here we test the deployed q-only observer under a changed physical condition: after burn-in, the Windy Pendulum damping profile switches from $D(\theta) = 0.3 + 0.5|\sin \theta|$ to $D(\theta) = 0.3 + 1.0|\sin \theta|$. All learned weights are fixed. The only online operation in PHAST-REOBSERVE is to re-run the causal observer/canonicalizer on the latest q-history window before applying the same pH transition. No parameters are updated online; the test isolates state re-estimation under a changed physical regime. The information contracts differ by row: PHAST and recurrent baselines consume q-histories, PHAST-REOBSERVE re-runs the observer online, finite-difference rows use reconstructed velocity/momentum, and template-known rows receive the exact 1-DOF plant form.

Table 7: **Dynamic damping-shift stress test.** Windy Pendulum, 5 seeds, q-only deployment except rows explicitly marked as finite-difference/template-known or oracle diagnostics. The shift occurs at step 70 after a 20-step burn-in. Lower is better for all metrics except recovery latency, where “–” means the method does not return below the configured threshold.

Method	Post MSE	CVaR90	Latency	Energy resid.	Power MAE	p RMSE
PHAST fixed	$2.30 \times 10^{-1} \pm 1.38 \times 10^{-1}$	$6.94 \times 10^{-1} \pm 4.21 \times 10^{-1}$	–	1.48 ± 0.77	0.426 ± 0.136	1.12 ± 0.55
PHAST reobserve	$1.30 \times 10^{-5} \pm 7.85 \times 10^{-6}$	$6.45 \times 10^{-5} \pm 4.48 \times 10^{-5}$	0	0.947 ± 0.470	0.225 ± 0.025	0.0556 ± 0.0233
PHAST oracle state	$3.16 \times 10^{-6} \pm 2.03 \times 10^{-6}$	$9.26 \times 10^{-6} \pm 5.72 \times 10^{-6}$	0	0.960 ± 0.471	0.213 ± 0.026	0
GRU	$1.95 \times 10^{-1} \pm 2.22 \times 10^{-2}$	1.45 ± 0.08	–	761.9 ± 99.2	–	–
Latent ODE/GRU	$5.24 \times 10^{-3} \pm 9.69 \times 10^{-4}$	$4.64 \times 10^{-2} \pm 7.86 \times 10^{-3}$	–	17.1 ± 4.1	–	–
HNN-FD	1.82×10^{-5}	7.92×10^{-5}	0	0.659	0.648	0.0829
pHNN-FD	1.68×10^{-5}	6.86×10^{-5}	0	0.167	0.146	0.0829
FD known	1.68×10^{-5}	6.86×10^{-5}	0	0.167	0.146	0.0829
EKF known	8.22×10^{-5}	5.16×10^{-4}	0	0.170	0.122	0.0528
RLS damping	1.68×10^{-5}	6.82×10^{-5}	0	0.148	0.127	0.0829

Table 7 isolates the reactive-observer mechanism. Re-observation improves post-shift q-error by four orders of magnitude over fixed-latent PHAST and is substantially better than the learned recurrent

619 and latent-ODE baselines. The oracle-state row shows that most of the PHAST gap is observer/state
620 estimation rather than the pH transition itself. Exact finite-difference, EKF, RLS, and template-known
621 pH estimators remain strong in this 1-DOF setting because they receive the plant structure explicitly.

622 **Mixed-coordinate Cart-Pole check.** The same stress test was repeated on Windy Cart-Pole, where
623 $q = (x, \theta)$ mixes a Euclidean cart coordinate with a periodic pole angle. This is a charted-observer
624 check rather than a new physical-accounting claim: if the observer finite-differences raw angles, a
625 branch-cut transition near $\pi / -\pi$ creates an artificial velocity of order $2\pi/\Delta t$. With $\Delta t = 0.02$, the
626 largest raw angle finite difference on the shifted test set was 313.6, while the wrapped finite difference
627 was 7.63. The observer therefore uses the product-coordinate chart $[x : \text{linear}, \theta : S^1]$. With this
628 charted observer, the partial-core Cart-Pole run has zero nonfinite predictions and preserves the same
629 reactive-observer pattern as the scalar pendulum.

Table 8: **Compact dynamic-shift summary.** Both rows keep learned weights fixed and re-run only the q-history observer/canonicalizer for PHAST re-observation. Windy Pendulum is the 5-seed scalar damping-shift result from Table 7; Windy Cart-Pole is a 3-seed mixed-coordinate check with strict nonfinite accounting. The best learned baseline is Latent-ODE/GRU for Windy Pendulum and a charted receding-window MLP for Windy Cart-Pole. The template ceiling uses finite-difference or oracle-template information and is not a q-history learned baseline.

Setting	PHAST fixed	PHAST reobserve	Best learned baseline	Template ceiling	Nonfinite rate
Windy Pendulum	2.30×10^{-1}	1.30×10^{-5}	5.24×10^{-3}	1.68×10^{-5}	0
Windy Cart-Pole, $q = (x, \theta)$	3.07×10^{-2}	1.64×10^{-6}	4.99×10^{-4}	1.41×10^{-6}	0

630 The Cart-Pole row sharpens the interpretation of this stress test. The gain is not merely from moving
631 to a higher-dimensional system; it comes from executing the observer on the correct coordinate
632 interface. Re-observation improves partial-core PHAST by roughly 1.9×10^4 over a fixed latent state,
633 by 3.3×10^5 over GRU, and by about 6.8×10^2 over Latent-ODE/GRU on post-shift MSE. It remains
634 about 3.0×10^2 lower in post-shift MSE than the strongest charted receding-window learned baseline,
635 an MLP that re-encodes the latest $[x, \sin \theta, \cos \theta]$ history at every step. The result supports the
636 charted state-estimation interface under a branch-cut sensitive regime change. Its role here is charted
637 state estimation rather than Cart-Pole damping or power-recovery superiority over template-known
638 estimators.

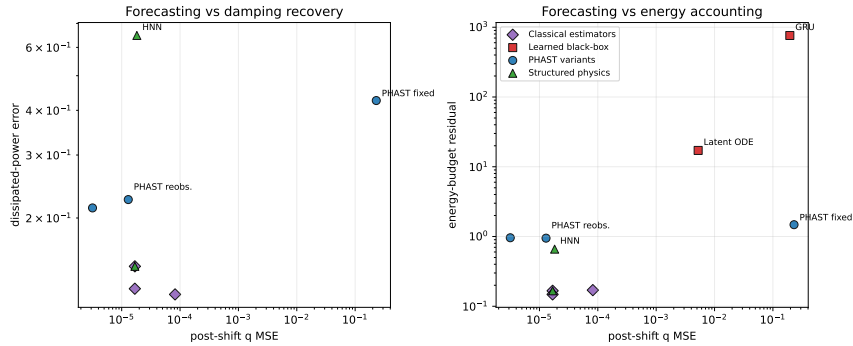


Figure 3: **Forecasting versus physical recovery after a damping shift.** The Pareto view separates post-shift q-error from physical accounting. HNN-FD remains competitive on q-error but has poor dissipated-power recovery; template-known estimators are strong when the exact 1-DOF plant form is supplied. PHAST re-observation is evaluated here as a q-only state-estimation mechanism under changed damping; template-known rows calibrate the difficulty of the 1-DOF setting.

639 J Symbol reference

640 Table 9 groups the symbols used in the main method and algorithms by the section in which they first
641 appear. Gray rows mark group headers.

Table 9: **Symbol reference.** Symbols are grouped by originating method component.

Symbol	Type / domain	Meaning
Problem and rollout contract		<i>Sec. 1</i>
q_t	$\mathcal{Q}$	Observed generalized position at time t
p_t	$\mathbb{R}^n$	Latent momentum or momentum-like phase variable
$x_t = (q_t, p_t)$	phase state	Canonical phase state used by the pH transition in the identity chart
K	$\mathbb{N}$	Burn-in context length for q-only histories
H	$\mathbb{N}$	Forecast or rollout horizon
$\hat{q}_{t+1:t+H}$	$\mathcal{Q}^H$	Autoregressive predicted positions
Charts, observer, and canonicalizer		<i>Sec. 3–3.2</i>
χ, χ^{-1}	$\mathcal{Q} \leftrightarrow \mathcal{Z}$	Domain chart and decoder; $z = \chi(q)$
z_t	$\mathcal{Z}$	Charted observation used by the observer and pH core
Π_z	phase state $\rightarrow \mathcal{Z}$	Projection from phase state to charted configuration coordinates
S_η	causal map	Baseline smoother or differentiator on charted histories
$\tilde{v}_t$	$T_{z_t} \mathcal{Z}$	Baseline chart-velocity estimate from S_η
r_ϕ	causal map	Neural residual correction to the baseline chart velocity
o_ϕ	causal map	Observer mapping a q-history to an estimated chart velocity
$\hat{z}_t$	$T_{z_t} \mathcal{Z}$	Observer-estimated chart velocity
C_ψ	$\mathcal{Z} \times T\mathcal{Z} \rightarrow \text{phase state}$	Canonicalizer mapping $(z_t, \hat{z}_t)$ to $(z_t, \hat{p}_t)$
σ_q	$\mathbb{R}_{>0}$	Observation-noise scale used by MAP/residual observer variants
Port-Hamiltonian core		<i>Sec. 3.1</i>
θ	parameter collection	Trainable parameters of typed physical components such as $V_\theta, M_\theta, D_\theta$
H_θ	scalar storage	Hamiltonian / stored energy
V_θ	scalar potential	Potential-energy component of H_θ
$M_\theta(q)$	SPD matrix	Mass or storage geometry converting momentum to velocity
J	skew matrix	Conservative exchange/interconnection bracket, $J = -J^\top$
R_θ	PSD matrix	Full dissipative metric channel in the pH vector field
G, u	port matrix, input	Work port and external input
y^{port}	port output	Power-conjugate output $G^\top \nabla H_\theta(x)$
Damping parameterization and spectra		<i>Sec. 3.3; App. H</i>
$D_\theta(q)$	PSD matrix	Mechanical damping channel acting on velocity
d_0	$\mathbb{R}_{\geq 0}$	Isotropic baseline damping floor
r	$\mathbb{N}$	Rank / number of anisotropic damping directions
$\beta_i(q)$	$\mathbb{R}_{\geq 0}$	Strength of low-rank damping direction i
$k_i(q)$	unit vector	Direction of rank-one damping term, $\ k_i(q)\ = 1$
$\bar{\beta}$	$\mathbb{R}_{\geq 0}$	Spectral/total damping cap, $\sum_i \beta_i(q) \leq \bar{\beta}$
$F(q)$	$\mathbb{R}^{n \times r}$	Orthonormal frame in the optional ordered partial-flag damping form
$\alpha_i(q)$	$\mathbb{R}_{\geq 0}$	Ordered modal damping strengths, $\alpha_1 \geq \dots \geq \alpha_r$
D_0	PSD matrix	Optional calibrated/modal damping anchor used for initialization
Integration and training		<i>Sec. 3.4</i>
Δt	$\mathbb{R}_{>0}$	Time step
$\Phi_{\Delta t}$	phase state $\rightarrow$ phase state	Full PHAST transition map; optionally a composition of L internal substeps
Φ_H	phase state $\rightarrow$ phase state	Conservative/symplectic substep
Φ_D	phase state $\rightarrow$ phase state	Dissipative half-step
L	$\mathbb{N}$	Number of internal PHAST substeps per observed transition
δt_s	$\mathbb{R}_{>0}$	Internal substep size for $s = 1, \dots, L$; fixed to $\Delta t/L$ in Table 5
B, T	integers	Minibatch size and trajectory length in training losses
$\mathcal{I}$	index set	Valid teacher-forced minibatch/time pairs
$N_{\mathcal{I}}$	integer	Number of teacher-forced index pairs, $ \mathcal{I} $
$\mathcal{L}_q$	scalar loss	Teacher-forced q-only next-step loss
$\mathcal{L}_p, \mathcal{L}_{\text{flow}}$	scalar losses	Optional momentum and chart-space flow supervision
$\mathcal{L}_{\text{port}}$	scalar loss	Optional conjugate port-output supervision
$\mathcal{L}_{\text{pass}}, \mathcal{L}_{\text{energy}}$	scalar losses	Optional passivity-violation and energy-budget terms
$\mathcal{L}_{\text{roll}}$	scalar loss	Optional open-loop autoregressive rollout loss
$\mathcal{L}_{\text{aux}}$	scalar loss	Weighted sum of optional non- $\mathcal{L}_q$ terms in Algorithm 3
$\mathcal{R}$	index set	Sampled rollout starts for $\mathcal{L}_{\text{roll}}$

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